

CPS4004 - Database Systems - Assessment Brief

Module Code:	CPS4004
Module Title:	Database Systems
Module Convenor:	Elisabetta Canetta
Module Level:	4

Assessment Number:	2
Assessment Title:	Northshore Logistic Ltd
Assessment Weight:	60%
Assessment Individual/Group:	Individual
Assessment Type:	Software Artefact with Report
Assessment Time/Word Count Restrictions:	1 software artefact 1800 words report
Assessment Time/Word Count Limit Consequences:	It is essential that assignments keep within the time/word count limit stated above. Any work beyond the maximum time/word length permitted will be disregarded and not accounted for in the final grade.

Issue Date:	03 th March 2026
Hand in Date:	06 th May 2026
Planned Feedback Date:	Within 3 working weeks
Mode of Submission:	Online via Moodle
Number of copies to be submitted:	1 copy of each of the following: - a report in pdf format - a zip file containing your software artefact

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Moderation Date:	07 th February 2026

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Introduction

This assessment aims to evaluate your ability to apply database concepts and technologies in a real-world scenario. You are tasked with designing and implementing a robust database solution for a fictional logistic company, Northshore Logistic Ltd. This organisation operates in a highly dynamic environment, handling vast amounts of data related to daily shipments, vehicle schedules, tracks driver routes, and stores customer order information.

Problem Scenario

Northshore Logistics Ltd is a medium-sized distribution company that operates warehouses and delivery hubs across the country. The organisation coordinates thousands of daily shipments, manages vehicle schedules, tracks driver routes, and stores customer order information. However, the current operations rely on a mix of paper-based logs, email communication, and separate spreadsheets maintained independently by each warehouse.

One major challenge faced by Northshore Logistics Ltd is the absence of an integrated database system to manage orders, inventory levels, vehicle assignments, and delivery records. Because each warehouse maintains its own files, the company lacks real-time visibility into stock availability, shipment statuses, and vehicle usage. This leads to frequent problems such as duplicated orders, incorrect stock counts, and inconsistent delivery timelines. The delivery process is especially affected. Drivers often receive outdated route sheets, causing delays, missed deliveries, or inefficient routing. Additionally, customer complaints about lost or delayed items are difficult to resolve because there is no central record tracking the location and status of each shipment at different stages of distribution.

Manual data entry increases the risk of human error, particularly when transferring information from paper forms into spreadsheets. This also slows down the reporting process, making it difficult for managers to monitor daily operations, identify delays, and plan staffing levels effectively.

Furthermore, the use of multiple unprotected files across different locations raises concerns about data security. Sensitive information, including customer addresses, payment records, and shipment details, is stored without proper access control, leaving the organisation vulnerable to data loss, unauthorised access, and accidental deletion.

Northshore Logistics Ltd now aims to implement a centralised database system to streamline operations, reduce errors, support secure data management, and ensure efficient coordination between warehouses, drivers, and customer service teams.

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Requirements

You are required to develop a database-driven system for Northshore Logistics Ltd. The requirements for the system are as follows:

- a) **Centralised Database System** - Develop a centralised database system using Python and SQLite to manage operations across all Northshore Logistics Ltd warehouses and delivery hubs. The system should consolidate information related to shipments, inventory levels, vehicle assignments, driver schedules, and delivery statuses. It should also generate operational insights such as delivery progress reports, warehouse activity logs, and vehicle utilisation summaries.
- b) **Shipment and delivery management features** - Implement comprehensive shipment management features, including:
 - Adding new shipment records (e.g., shipment ID, order number, sender/receiver details, item description);
 - Updating delivery information (e.g., delivery date, assigned driver, route details);
 - Tracking incident reports (e.g., delays, route changes, damaged goods, failed delivery attempts);
 - Managing financial details related to shipments (e.g., transportation cost, surcharges, payment status);
 - Generating reports on shipment status and resolution (e.g., delivered, delayed, returned to warehouse, in transit).
- c) **Inventory and fleet management** - Integrate inventory and fleet management functionalities into the system to support daily warehouse operations:
 - Track inventory levels across multiple warehouses (e.g., stock quantity, reorder levels, item location);
 - Manage vehicle information (e.g., vehicle ID, capacity, maintenance schedule, availability);
 - Record and update driver information (e.g., licenses, route history, shift assignments);
 - Monitor warehouse activity, including inbound/outbound shipments, restocking, and item transfers between locations;

These features should help optimise stock handling, warehouse efficiency, and delivery coordination.

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- d) **Data security measures:** Implement strong data security measures to protect sensitive operational information stored in the centralised database. This includes:
- Role-based access control (e.g., warehouse staff, drivers, managers, admin roles);
 - Encryption of sensitive data, such as customer addresses, payment records, and driver personal information;
 - Logging and auditing of database activities to detect suspicious access, unauthorised modifications, or data misuse.
- e) **Scalability and performance:** Design the database system with scalability in mind to accommodate the growing data needs of Northshore Logistics Ltd. The system should be able to:
- Handle increasing numbers of shipments, drivers, vehicles, and warehouses;
 - Maintain good performance during peak periods (e.g., seasonal demand).
- f) **User-friendly interface:** Develop a simple, intuitive user interface for employees across warehouses and delivery hubs. The interface should allow users to easily:
- Add, update, and retrieve shipment records;
 - Track delivery progress and vehicle deployment;
 - Record inventory changes and view stock levels;
 - Access important operational summaries and reports.

The interface should be suitable for staff with varying technical abilities, helping improve overall efficiency and reduce data entry errors.

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Environment and Tools

You are required to use any of the following tools:

- **Adobe PDF Reader**: To view your final report which is saved as a PDF file.
 - **DBML**: As your mark-up language.
 - **Draw.io** (or equivalent tool): To create diagrams.
 - **Git Tools** and **GitHub** for version control
 - **Microsoft Word** (or equivalent tool): To write your report and export as a PDF.
 - **PyCharm** (or equivalent) as your integrated development environment
 - **Python 3.11+** as the standard python library
 - **SQLite 3** as your database management system
 - **SQLite Studio** (or equivalent tool) to browser your database
- Additionally, the following libraries/modules may be imported and utilised:
- pandas** - for data handling
 - datetime** - for handling dates/times
 - logging** - for logging information
 - hashlib** - for hashing passwords before storing them in the database
 - secrets** - for token generation for authentication
 - sqlite3** - to develop your database
 - tkinter** - to create a graphical interface
- No other python libraries or modules should be used

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Submission

The assessment must be completed individually. You must not share, in part or whole, your assessment with another party other than the module convenor and for the purpose of submission to the university. You must ensure that the University's academic misconduct guidelines are followed in their entirety.

You should use the assessment submission link on the module's Moodle page to submit the following files:

- A **PDF** file for your report. This should not be included in the zip file but instead submitted as a separate file. Failure to do so may result in zero being awarded.
- A **Zip** file of your software artefact. This should contain your solution files, git log, and other files required to view and run your solution.

You should ensure that you make a timely submission by the deadline stated at the start of this assessment brief.

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Assessment Criteria

Your assessment will be graded according to the following criteria:

Grading criteria	Functionality (25%)	Documentation (45%)	Professional Practice (30%)
Mark band			
80-100 Pass (1st)	Displays exceptional skills with comprehensive implementation and design considerations including strong considerations and implementation for efficiency, security, and scalability.	Exceptional analysis and evaluation provided, offering deep insights into the software artefact's strengths and weaknesses, supported by evidence and critical reflection. Report is exceptionally well-structured with use of referencing and citations, with thorough and accurate citation of relevant sources to support arguments and analysis.	Demonstrates exceptional adherence to industry and ethical practice with strong usage of version control with a well-structured repository, regular commits, detailed commit messages, and evidence of branching and merging strategies. Ethical considerations are very clearly discussed.
70-79 Pass (1st)	Exhibits advanced skills with comprehensive database development, and in-depth design considerations. Implements features for efficiency, security, and scalability.	Advanced communication in explaining complex software development concepts. Well-structured report with appropriate citations and captions where relevant.	Advanced adherence to code conventions, ethics, and relevant industry practice with usage of version control with a well-structured repository, regular commits, detailed commit messages, and some evidence of branching and merging strategies. Ethical considerations are clearly discussed.
60-69 Pass (2.1)	Good, consistent knowledge and understanding of the material, main concepts, key theories, and practice at this level. Exhibits advanced skills in database application development.	Strong communication in explaining complex programming and problem-solving concepts.	Competent adherence to ethics, code conventions and other relevant industry practice with usage of version control with regular commits and meaningful commit messages that provide clear context for changes. Ethical considerations are well discussed.
50-59 Pass (2.2)	Sound, routine knowledge and understanding of the material, main concepts and key theories. Some flaws may be evident. Shows improved skills with refined implementation.	Improved communication skills seen in summarising design and implementation.	Improved adherence to code conventions and other relevant industry practice. Some ethical considerations. Usage of version control with regular commits. Commit messages provide some context but may lack detail or consistency. Ethical considerations are discussed satisfactorily.
40-49 Pass (3rd) (Threshold)	Demonstrates basic skills in developing a functional application	Basic communication evident in summary of implemented solution.	Basic adherence to code conventions and other relevant industry practice with evidence of version control usage. Commits are sporadic and lack meaningful commit messages. Ethical considerations are discussed sufficiently.
30-39 Fail	Displays limited proficiency, with significant deficiencies in the programming.	Demonstrates inadequate communication skills, struggling to articulate ideas and concepts.	Inadequate adherence to code conventions and other relevant industry practice. Minimal evidence of version control. Ethical considerations are discussed poorly.
0-29 Fail	Demonstrates severely inadequate skills, unable to create a basic functional program.	Poor communication skills, unable to provide clear summary.	Poor adherence to code conventions and other relevant industry practice. Little or no evidence of version control. Ethical considerations are not discussed.

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Learning Outcomes

This assessment will enable students to demonstrate the following learning outcomes as stated in the module outline:

[Module Learning Outcome 2]

- Demonstrate a comprehensive understanding and professional practice of utilising databases in diverse contexts, such as web and mobile applications, data warehousing, and business intelligence, while considering their ethical implications.

How is this learning outcome addressed?

The assessment requires designing and implementing a centralised database system for an insurance company, reflecting practical understanding of database usage in real-world contexts. Integrating database functionalities for claims processing and policy management addresses diverse applications. Implementing data security measures demonstrates consideration for ethical implications.

[Module Learning Outcome 3]

- Apply normalisation and denormalisation techniques to design efficient and effective databases.

How is this learning outcome addressed?

Individuals apply normalisation and denormalisation techniques to design an efficient database system for the insurance company. This aligns with the learning outcome by ensuring the database schema minimises redundancy and dependency issues while enhancing query performance as needed.

[Module Learning Outcome 4]

- Critically evaluate the performance, security and privacy issues related to database systems.

How is this learning outcome addressed?

The assessment mandates implementing data security measures and evaluating system performance, addressing security, privacy, and performance concerns. This aligns with the learning outcome by necessitating critical evaluation of these aspects in the context of database systems.

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[Module Learning Outcome 6]

- Create and deploy a database system for a practical application, considering ethical implications and adherence to professional standards throughout the design and implementation process.

How is this learning outcome addressed?

The assessment involves creating and deploying a practical database system, considering ethical implications throughout the process. Adherence to professional standards is implicit in developing a robust solution for the insurance company.

[Module Learning Outcome 7]

- Independently plan and execute a computer-based system project, utilising effective problem-solving and information management skills, numeracy, communication skills, and a commitment to ongoing professional development.

How is this learning outcome addressed?

Individuals plan and execute the database system project, demonstrating problem-solving, information management, numeracy, and communication skills. The commitment to ongoing professional development is encouraged through practical application in claims processing and policy management.

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Regulations, Policies, and Guidelines

Guidance for online submissions

<https://www.stmarys.ac.uk/policies/online-submissions.aspx>

Academic Misconduct

Any submission must be students' own work and, where facts or ideas have been used from other sources, these sources must be appropriately referenced. Please find a link to the academic misconduct policy below:

<https://www.stmarys.ac.uk/policies/academic-regulations.aspx>

Ethics Policy

The work being carried out by students must be in compliance with the Ethics Policy. Where there is an ethical issue, as specified within the Ethics Policy, then students will need ethical approval prior to the start of the project. Please find a link to the ethics policy below:

<https://www.stmarys.ac.uk/research/students/ethical-review-process.aspx>

Extenuating Circumstances

The University's Extenuating Circumstances procedure helps students facing challenges in assessment submission. To request an extension or deferment, submit an EC application with evidence. Approved cases will not incur academic penalties. For longer-term issues, contact Student Services. Please find a link to the EC policy below:

<https://www.stmarys.ac.uk/policies/extenuating-circumstances.aspx>